

Unit 8

REACTION KINETICS

21 items · 57 marks

Syllabus sections in this unit

8.1 Rate of reaction

8.2 Effect of temperature on reaction rates and the concept of activation energy

8.3 Homogeneous and heterogeneous catalysts

Unit 8: Reaction kinetics

1. M/J 26/P21/Q2(b)(ii)

Context

2 The Contact process uses sulfur to produce sulfuric acid.

stage 1 Sulfur is burned in excess air.

stage 2 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -196 \text{ kJ mol}^{-1}$

stage 3 SO_3 reacts to produce H_2SO_4 .

Context: part (b) (introduction)

(b) The reaction conditions for stage 2 include:

- using a ratio of 1 mol SO_2 to 1 mol O_2
- heat in the presence of V_2O_5 .

(ii) Explain how V_2O_5 affects the rate of reaction.

[2]

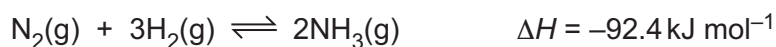
2. M/J 26/P22/Q1(e)(ii)

Context

1 Oxygen reacts with many elements.

Context: part (e) (introduction)

(e) Ammonia is made in the Haber process.



A temperature of 400–450 °C is used in the Haber process.

(ii) Describe how the initial rate of the forward reaction changes, if at all, when a temperature greater than 450 °C is used. Explain your answer.

Assume all other variables remain constant.

change in initial rate

explanation

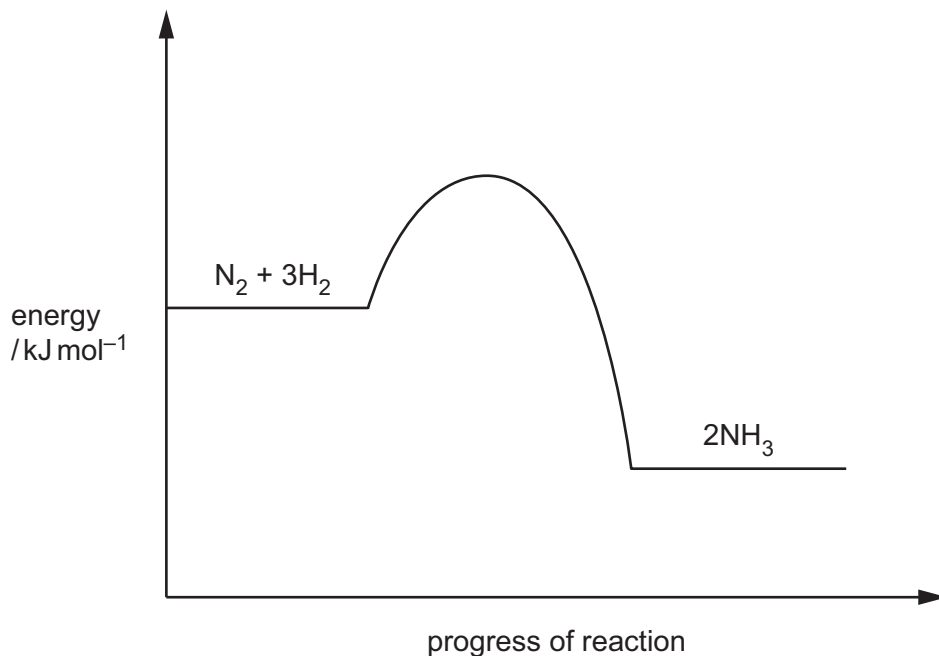
[3]

3. M/J 26/P24/Q2(b)(i)

Context: part (b) (introduction)

(b) Nitrogen is used to manufacture ammonia, NH_3 , in the Haber process. An iron catalyst is used.

(i) Complete the reaction pathway diagram in Figure 2.1 to show the effect of adding an iron catalyst to the reaction of nitrogen and hydrogen.

**Figure 2.1**

[1]

4. M/J 26/P24/Q2(d)

(d) Explain the effect of increasing temperature on rate of reaction.

[3]

5. M/J 26/P24/Q2(e)

(e) Figure 2.2 shows the Boltzmann distribution for a sample of X(g) at temperature T_1 .

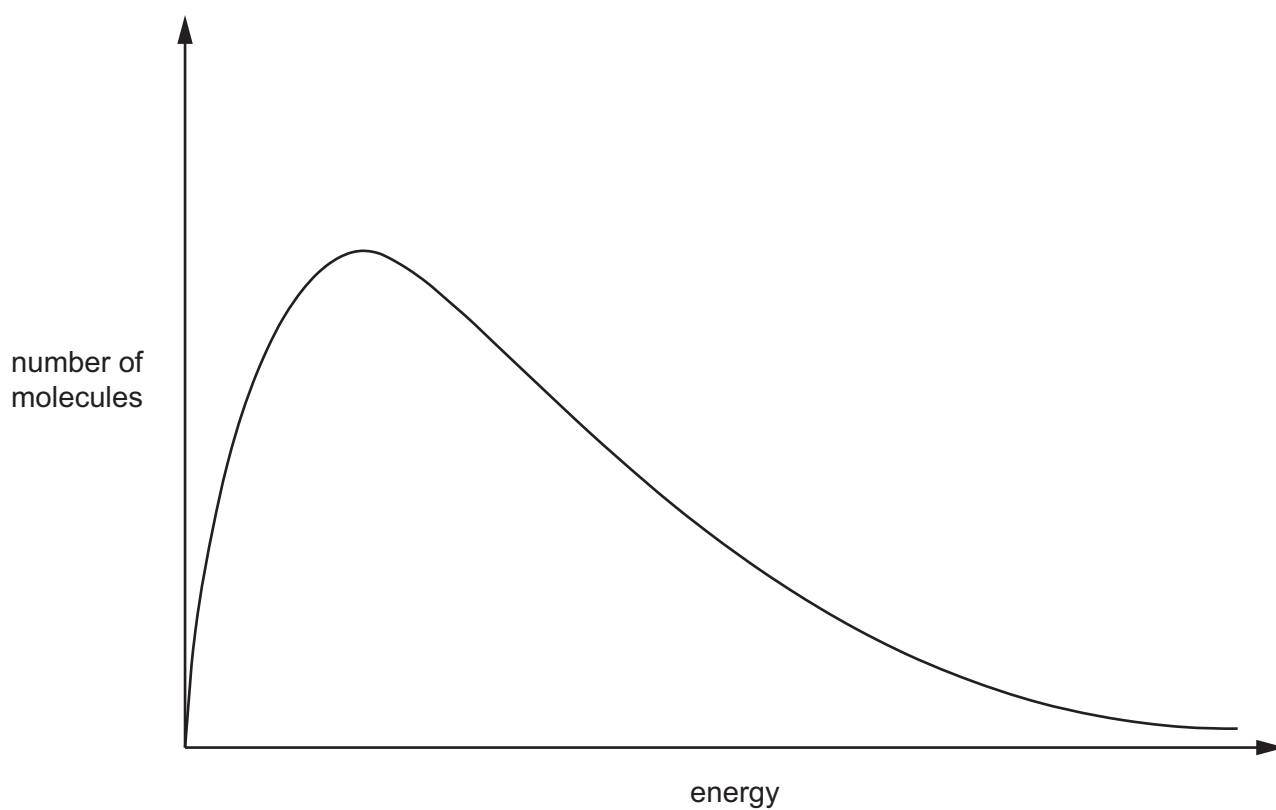


Figure 2.2

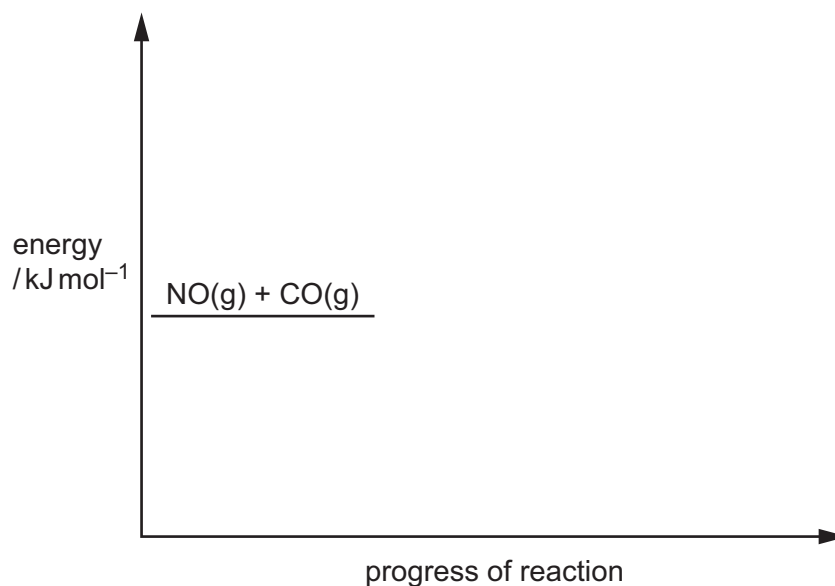
Sketch a curve on Figure 2.2 to show the Boltzmann distribution for the sample of X(g) at a temperature higher than T_1 . [2]

6. MAR 26/P22/Q2(d)(ii)

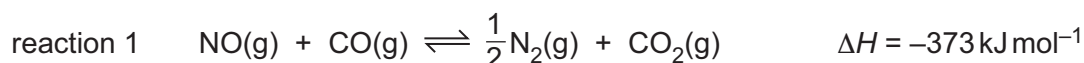
Context

2 Nitrogen monoxide, NO, is an atmospheric pollutant.

Context: Fig. 2.3

**Figure 2.3**

Context: part (d) (introduction)

(d) Reaction 1 occurs during the catalytic removal of NO from car exhaust gases.The activation energy, E_a , of this reaction is 570 kJ mol^{-1} .**(ii)** On your diagram in Figure 2.3, show how the addition of a catalyst affects reaction 1. [1]**7. O/N 25/P24/Q2(c)(i)**

Context

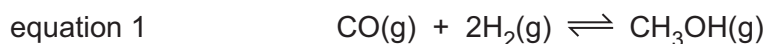
2 Aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , are both used as reagents and catalysts.

Context: part (b) (introduction)

(b) A 5.00 dm^3 sealed flask contains 0.400 mol of CO(g) and 0.800 mol of $\text{H}_2\text{(g)}$ and an Al_2O_3 catalyst. The flask is heated to a temperature of 290°C and allowed to reach equilibrium. Equation 1 shows the reaction.The equilibrium constant, K_c , of equation 1 is given.

$$K_c = \frac{[\text{CH}_3\text{OH}]}{[\text{CO}][\text{H}_2]^2}$$

Context: part (c) (introduction)

(c) P_4O_{10} catalyses the reversible reaction of CO with H_2 to form CH_3OH . P_4O_{10} then acts as a dehydrating agent, causing CH_3OH to form CH_3OCH_3 .**(i)** Explain how the presence of a catalyst affects a chemical reaction.

[1]

8. M/J 25/P21/Q3(a)

- 3 (a) Hexene reacts with hydrogen gas to produce hexane.

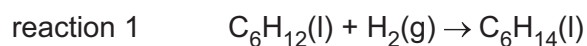


Fig. 3.1 shows the distribution of energies of $\text{H}_2(\text{g})$ molecules at temperature T .

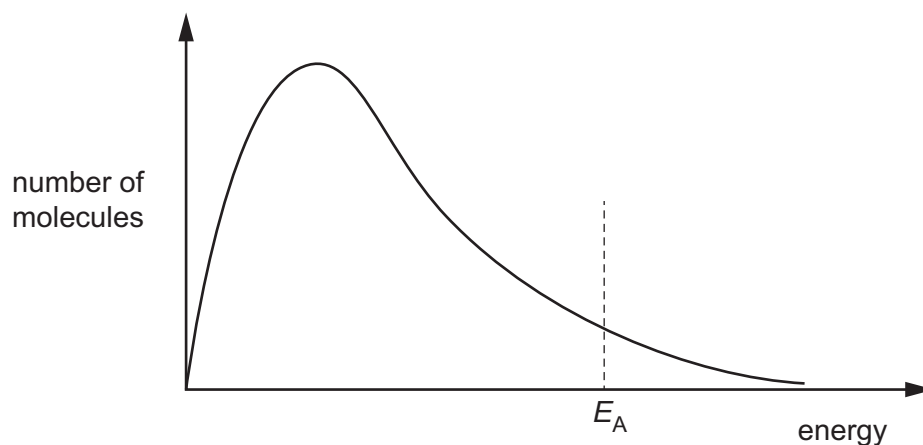


Fig. 3.1

- (i) Sketch on Fig. 3.1 the shape of the curve for the same sample of $\text{H}_2(\text{g})$ molecules when the temperature is increased. [2]
- (ii) Explain why increasing the temperature increases the rate of reaction 1. [2]
- (iii) State the role of nickel when it is added to reaction 1. [1]
- (iv) Annotate Fig. 3.2 to show the effect of adding nickel to reaction 1 at temperature T .

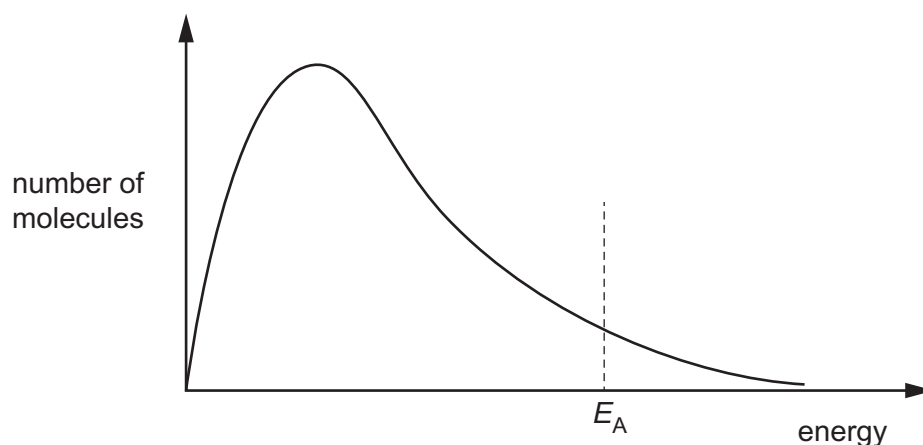


Fig. 3.2

[1]

9. M/J 25/P22/Q3(b)

- (b) $\text{H}_2(\text{g})$ and nickel are added to alkene **X**. Fig. 3.2 shows how the concentration of **X** changes with time.

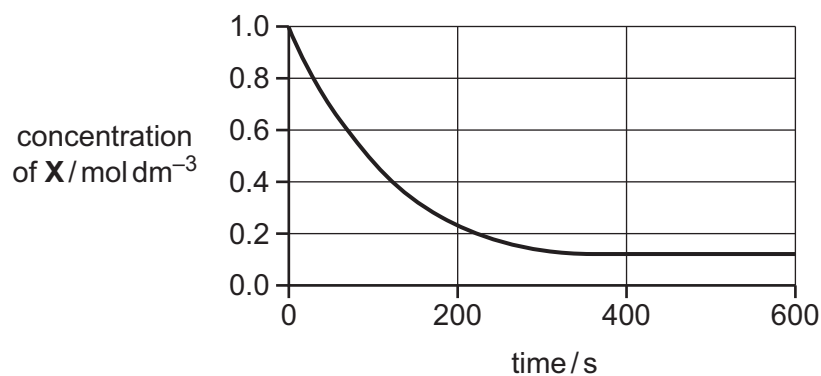


Fig. 3.2

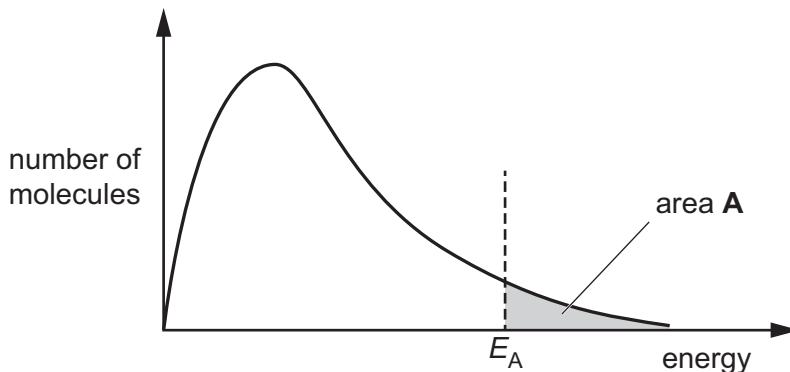
- (i) Use Fig. 3.2 to identify the limiting reagent in this reaction. Explain your answer. [1]
- (ii) Use Fig. 3.2 to describe how the gradient changes as the reaction proceeds. State what this shows about the rate during this reaction. [1]

10. M/J 25/P22/Q3(d)

Context: part (c)

(c) Hydrocarbon **Y** contains two C=C groups. There are no other functional groups present.**Y** reacts with an excess of $\text{H}_2(\text{g})$ to produce hexane, C_6H_{14} .**Table 3.1**

bond	energy / kJ mol^{-1}
C–C	350
C=C	610
$\text{C}\equiv\text{C}$	840
H–H	436
C–H	410

Use Table 3.1 to calculate the enthalpy change per mole of C_6H_{14} produced in this reaction.enthalpy change = kJ mol^{-1} [2]**(d)** Hydrocarbon **Y** reacts with $\text{H}_2(\text{g})$.Fig. 3.3 shows the distribution of energies of $\text{H}_2(\text{g})$ at temperature T .Area **A** represents the number of molecules with energy greater than or equal to the activation energy, E_{A} , at temperature T .**Fig. 3.3****(i)** Annotate Fig. 3.3 to show the effect of adding nickel to **Y** and $\text{H}_2(\text{g})$ at temperature T . [1]**(ii)** Area **B** (**not** labelled on Fig. 3.3) represents the number of molecules with energy greater than or equal to the activation energy when nickel is added at temperature T .State the difference, if any, between areas **A** and **B**. Explain the significance of your answer on the rate of hydrogenation of **Y**. Give your answer in terms of collisions.

[2]

11. M/J 25/P23/Q3(b)(ii)-(iii)

Context: part (b) (introduction)

(b) C_4H_8 reacts with an excess of H_2 to produce C_4H_{10} .



(ii) Define activation energy, E_{A} .

[1]

(iii) The Boltzmann distribution for the reaction mixture in reaction 1 is shown in Fig. 3.1. Use the Boltzmann distribution to explain the effect of adding a catalyst on the rate of reaction.

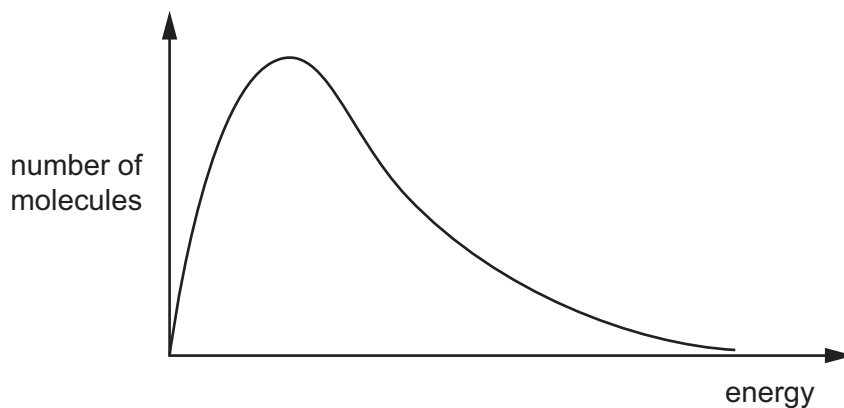


Fig. 3.1

[2]

12. M/J 25/P23/Q4(a)

Context

4 The reaction between $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ and $\text{HCl}(\text{aq})$ is monitored at constant temperature.

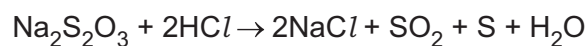


Fig. 4.1 shows how the concentration of $\text{HCl}(\text{aq})$ varies with time.

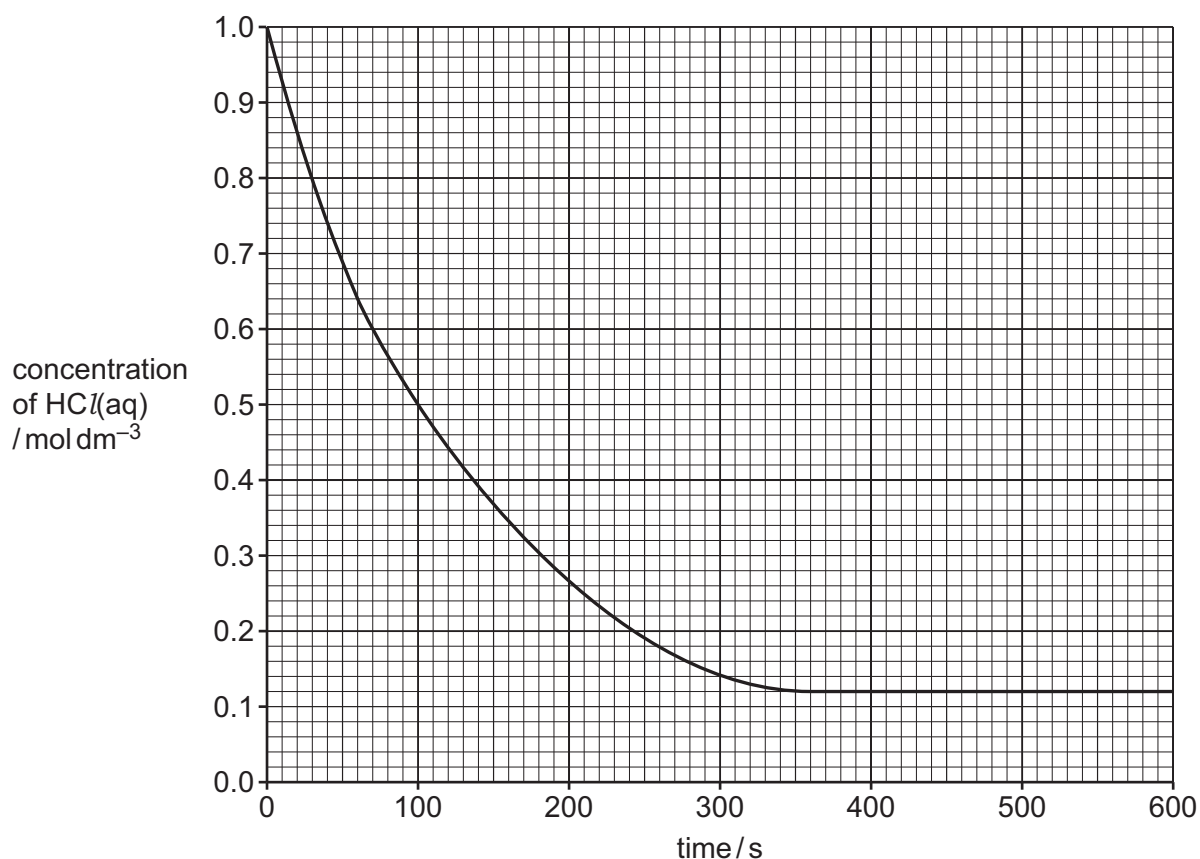


Fig. 4.1

- (a) (i) Use Fig. 4.1 to find the average rate of change of concentration of HCl(aq) in this reaction between 0–100 seconds and between 400–500 seconds. Include units in your answers.
- 0–100 seconds units
- 400–500 seconds units
- [2]
- (ii) Use Fig. 4.1 to identify the limiting reagent. Explain your answer.
- [1]
- (iii) Explain why the rate of reaction changes with time.
- [1]
- (iv) The reaction between Na₂S₂O₃(aq) and HCl(aq) is repeated in a second experiment.
- In this second experiment, 25.0 cm³ of 0.050 mol dm⁻³ Na₂S₂O₃(aq) reacts with 0.0020 mol of HCl(aq). Calculate the number of sulfur atoms produced.
- number of sulfur atoms produced = [2]
- (v) Explain why the rate of reaction **cannot** be monitored accurately by measuring the volume of SO₂(g) produced in this reaction.
- [1]

13. M/J 24/P23/Q2(b)(ii)

Context

- 2** $\text{N}_2(\text{g})$ reacts with $\text{H}_2(\text{g})$ in the Haber process, as shown in reaction 1.



Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	B	C
initial molar ratio of $\text{N}_2 : \text{H}_2$ added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $\text{NH}_3(\text{g})$ at equilibrium	58	x	y

- (ii) Describe and explain the change to the rate of the forward reaction that occurs to establish the equilibrium in **C** compared to **A**.

You do **not** need to refer to the Boltzmann distribution in your answer.

[2]

14. MAR 24/P22/Q3(c)(ii)-(iii)

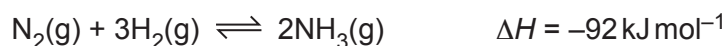
Context

- 3** Nitrogen, N_2 , is generally an unreactive molecule but it does react under certain conditions.

Context: part (c) (introduction)

- (c) The Haber process involves the reaction of N_2 and H_2 to form ammonia, NH_3 .

A catalyst is used, which allows the process to be carried out at a lower temperature and pressure.



- (ii) Explain how the presence of a catalyst affects the reaction.

[1]

- (iii) State and explain the effect, if any, on the **rate** of the Haber process as the pressure is lowered.

[2]

15. O/N 23/P21/Q3(b)

Context

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(b) $\text{KClO}_3(\text{s})$ decomposes when heated.

$\text{MnO}_2(\text{s})$ catalyses the exothermic decomposition reaction.

Complete and label the diagram in Fig. 3.1 to show the effect of $\text{MnO}_2(\text{s})$ on the decomposition of $\text{KClO}_3(\text{s})$.

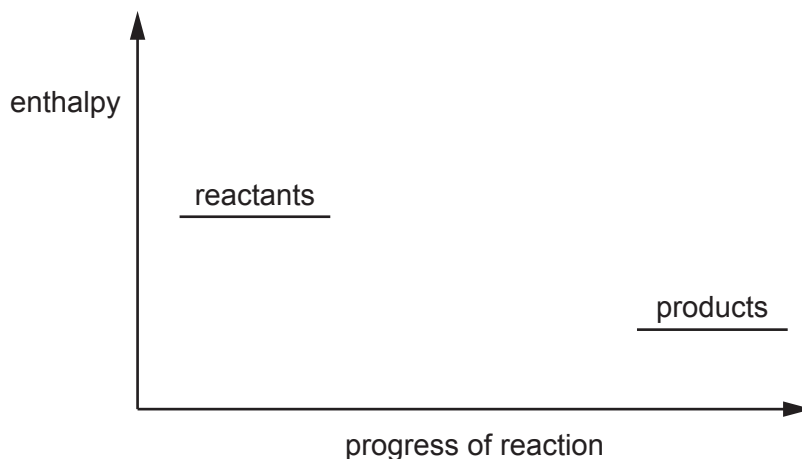


Fig. 3.1

[2]

16. O/N 23/P23/Q3(b)

Context

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(b) $\text{KClO}_3(\text{s})$ decomposes when heated.

$\text{MnO}_2(\text{s})$ catalyses the exothermic decomposition reaction.

Complete and label the diagram in Fig. 3.1 to show the effect of $\text{MnO}_2(\text{s})$ on the decomposition of $\text{KClO}_3(\text{s})$.

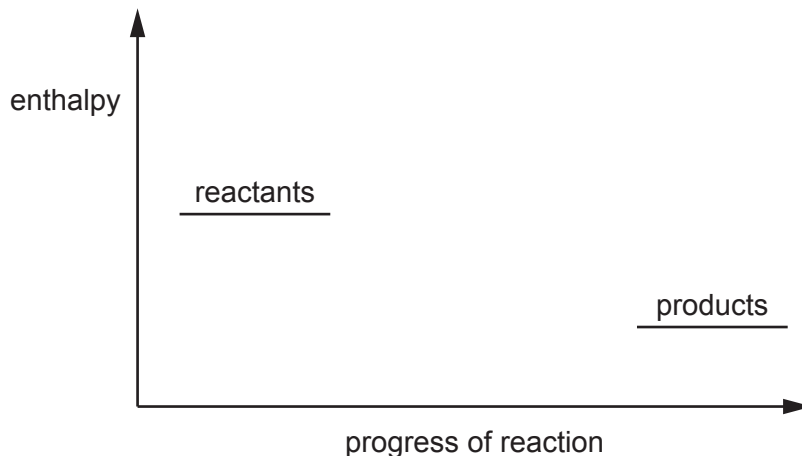


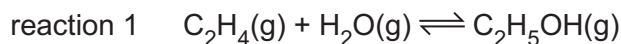
Fig. 3.1

[2]

17. M/J 23/P22/Q4(c)(iv)-(v)

Context

4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



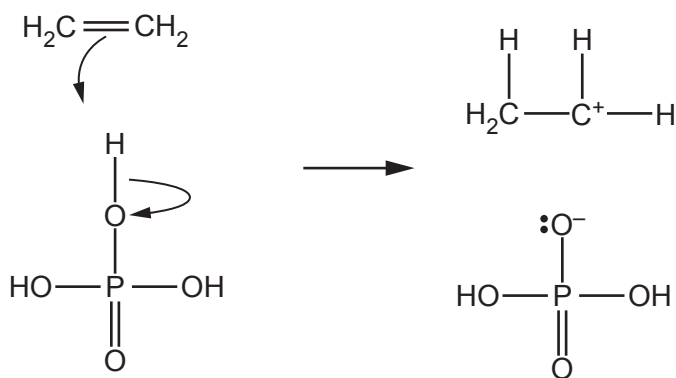
Context: Fig. 4.2

**Fig. 4.2**

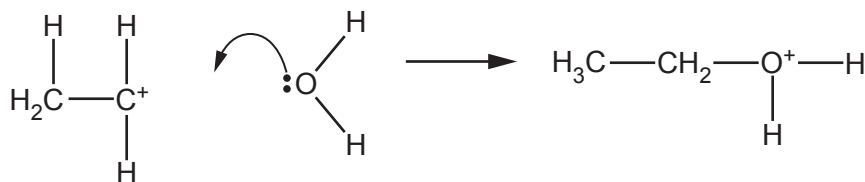
Context: part (c) (introduction)

(c) The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



step 2

**Fig. 4.1**

(iv) Describe how a catalyst affects a reaction. Explain your answer.

[2]

(v) Use Fig. 4.1 and Fig. 4.2 to justify why H_3PO_4 is described as a catalyst in reaction 1.

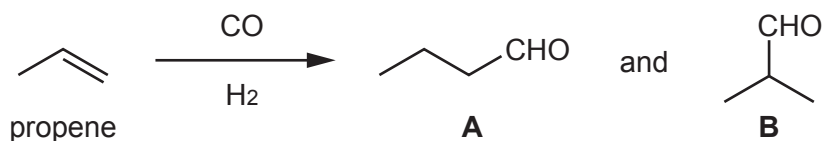
[1]

18. MAR 23/P22/Q3(d)(i)

Context

- 3** Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

**Fig. 3.1**

Context: part (d) (introduction)

- (d)** The reaction of ethene, C₂H₄, with a 1:1 mixture of CO and H₂ is shown in equation 1.



At atmospheric pressure a cobalt-based catalyst is used in this reaction.

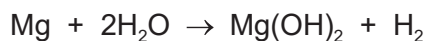
- (i)** State and explain the effect of using a catalyst on this reaction.

[2]

19. M/J 22/P21/Q1(g)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.
- (g)** Cold water reacts slowly with a piece of Mg to produce bubbles of H₂(g).
Cold water reacts rapidly with burning Mg to produce H₂(g) in an explosive mixture.

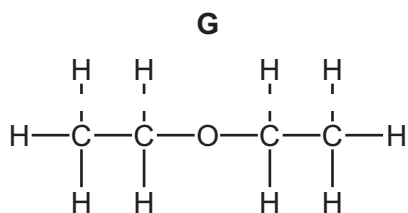


Explain why the rate of reaction of cold water with burning magnesium is greater.

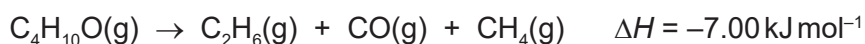
[2]

20. M/J 22/P23/Q3(a)(iv)-(v)

Context

3 G belongs to a group of compounds called ethers.**Fig. 3.1**

Context: part (a) (introduction)

(a) When **G**, $C_4H_{10}O$, is heated, thermal decomposition occurs.**(iv)** Thermal decomposition of **G** in the presence of I_2 affects the activation energy, E_a , for the reaction. Table 3.2 shows E_a for the thermal decomposition of **G** with and without I_2 .**Table 3.2**

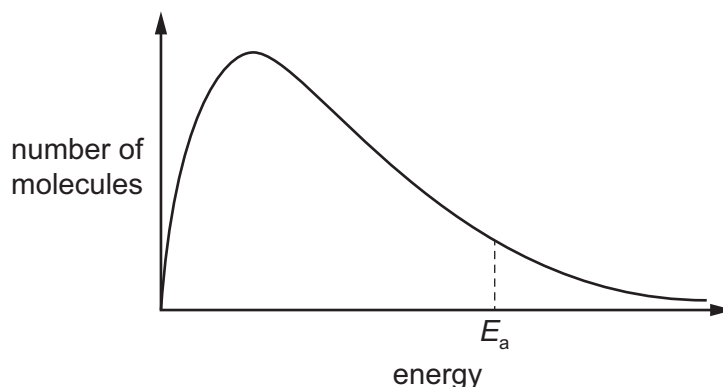
reaction	E_a (with I_2)/kJ mol ⁻¹	E_a /kJ mol ⁻¹
$C_4H_{10}O(g) \rightarrow C_2H_6(g) + CO(g) + CH_4(g)$	143	224

State what effect adding I_2 to the reaction mixture has on the value of K_c .
Explain your answer.

[2]

(v) Fig. 3.2 shows the Boltzmann distribution of energies for molecules of **G** at constant temperature, $T^\circ\text{C}$.

Sketch, on Fig. 3.2, the Boltzmann distribution of energies for molecules of **G** at a higher temperature, $(T+100)^\circ\text{C}$.

**Fig. 3.2**

[2]

21. MAR 22/P22/Q2(b)(iii)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (b) (introduction)

(b) Al_2O_3 is an amphoteric oxide found in bauxite.

(iii) Al_2O_3 is used as a catalyst in the dehydration of alcohols.

State the effect of using Al_2O_3 as a catalyst in the dehydration of alcohols. Use the Boltzmann distribution in Fig. 2.1 to help explain your answer.

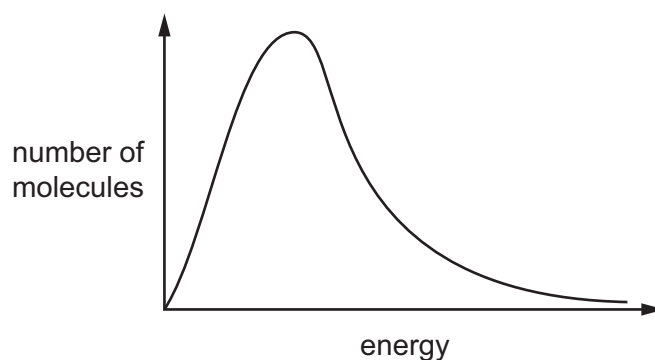


Fig. 2.1

[3]

Answers Section

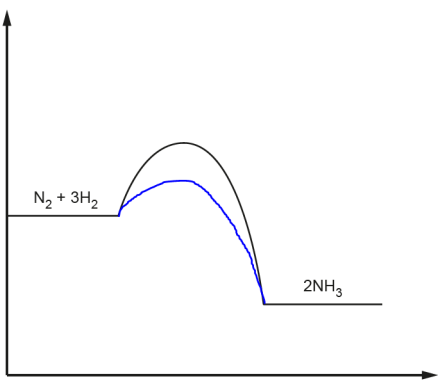
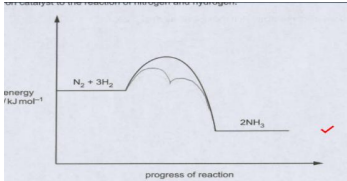
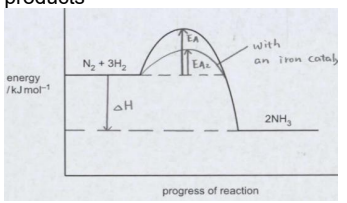
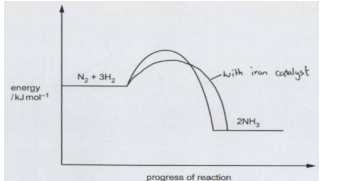
1. M/J 26/P21/Q2(b)(ii)

2(b)(ii)	M1 it / rate (of reaction) increases M2 (it) lowers EA OR (used) to lower EA	2 rate = speed successful = effective particles = molecules M1 A greater frequency of successful collisions / more successful collisions per unit time / higher successful collision rate / more frequent successful collisions / more successful collisions per second A for per = / M2 A more O₂ and/or SO₂ (molecules) OR more particles have sufficient / enough / minimum energy to react M2 A more O₂ and / or SO₂ (molecules) OR more particles have energy $\geq$ EA A Ea / EA / AE for EA I more of the collisions have energy greater than or equal to activation energy I pathway (even if wrong) I redox I it is a catalyst
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2. M/J 26/P22/Q1(e)(ii)

1(e)(ii)	change in rate explanation M1 increases M2 more particles have $E \geq E_A$ M3 so greater frequency of effective collisions.	3 I yield / Le Chatelier arguments M1 might be seen in explanation if not on first answer line M1 A it takes less time to reach equilibrium M2 A more particles have energy greater than / equal to the activation energy / E_A M2 R particles have more activation energy M2 I particles have more energy M2 I atoms / reactants M3 A more effective collisions per unit time / increased rate of effective collisions M3 I higher chance / probability of more successful collisions M3 I more collisions / more effective collisions alone effective = successful particles = molecules activation energy = sufficient / enough energy to react
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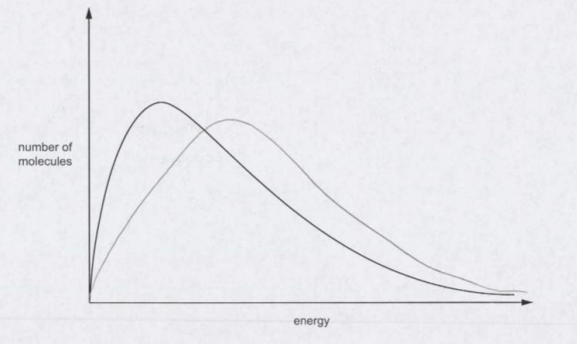
3. M/J 26/P24/Q2(b)(i)

2(b)(i)		1. A BOD double hump for cat complex intermediate  A line that joins original line before products  A BOD 
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4. M/J 26/P24/Q2(d)

2(d)	M1 (increasing temperature) increases rate M2 increased number of particles/collisions with E_A OR larger proportion of particles/collisions have energy greater than or equal to E_A / $E \geq E_A$ M3 increased / higher frequency of effective / successful collisions	3 M1 A BOD increases speed of reaction M2 A more particles with enough / sufficient energy to react M3 A increased rate of effective / successful collisions A increase in effective / successful collisions per unit time A 'increased frequency of collisions' AND 'more effective / successful collisions' I increased chance / probability of effective / successful collision I more collisions (only) I references to Le Chatelier / yield A particles = molecules = atoms
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5. M/J 26/P24/Q2(e)

2(e)	 • start at origin and not crossing original until beyond the peak • peak to the RHS of original peak AND lower than original • line does not cross the original after the peak reached AND crosses the original once only ✓✓	2 A BOD slight upwards 'kick' at the end. Max 1 completely correct graphs for temperature < T₁
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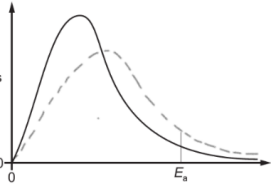
6. MAR 26/P22/Q2(d)(ii)

2(d)(ii)	diagram has lower-energy transition state(s)	1
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7. O/N 25/P24/Q2(c)(i)

2(c)(i)	increases rate by providing a different mechanism with a lower activation energy	1
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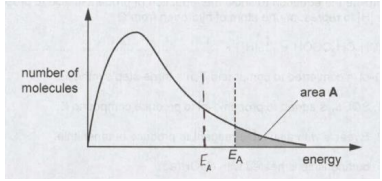
8. M/J 25/P21/Q3(a)

3(a)(i)	 Two correct for one mark, all four for two marks: • start at origin • peak sits below the original • peak sits to the right-hand side of the original • new line crosses the original once only AND to the right of the original peak	2
3(a)(ii)	M1 greater proportion of (H₂) molecules have energy greater than or equal to the activation energy / $\geq E_A$ M2 increase in frequency of successful collisions (between H₂ and C₆H₁₂)	2
3(a)(iii)	catalyst	1
3(a)(iv)	annotation to show new E _A (nickel / catalyst) further to the left than original E _A	1

9. M/J 25/P22/Q3(b)

3(b)(i)	H ₂ / hydrogen AND some X / alkene left(over) / remains (at the end of this reaction)	1
3(b)(ii)	link the significance of gradient values to rate AND describe either the decrease in rate or the decrease in gradient as the gradient decreases the rate decreases OR rate decreases until the reaction stops when gradient = 0	1

10. M/J 25/P22/Q3(d)

3(d)(i)	E_A shown on Fig. 3.3 to the left of original (on the x axis)	1
3(d)(ii)	 M1 it / (area) B larger AND rate (reaction / hydrogenation) increases OR (area) B identified by shading or labelling on Fig 3.3 AND rate (reaction / hydrogenation) increases in prose M2 increase in the frequency of effective / successful collisions OR increase in rate of effective / successful collisions	2

Answer for context part (c)

3(c)	method 1 – breaking all bonds in molecules $(2 \times 610) + (2 \times 436) + (10 \times 410) + (3 \times 350) - (14 \times 410) - (5 \times 350)$ - energy involved when all bonds are broken = (7242) - energy involved when all bonds are made = (7490) - energy to break bonds – energy to make bonds $= (7242 - 7490) = -248$ method 2 – ignore any bonds that do not change from reactants to products $(2 \times 610) + (2 \times 436) - (4 \times 410) - (2 \times 350)$ - energy involved when all bonds are broken = (2092) - energy involved when all bonds are made = (2340) - energy to break bonds – energy to make bonds $(2092 - 2340) = -248$	2
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11. M/J 25/P23/Q3(b)(ii)-(iii)

3(b)(ii)	minimum energy required for a collision to be effective	1
3(b)(iii)	M1 • area under the curve with energy $\geq E_A$ using a catalyst is larger so • greater proportion of particles with sufficient energy to react M2 rate faster because higher frequency of successful collisions	2

12. M/J 25/P23/Q4(a)

4(a)(i)	Check final graph for range of acceptable answers $\pm(1-0.5)/100 = \pm 0.005$ 0 $\text{mol dm}^{-3} \text{ s}^{-1}$ for both Any 2 points = 1 mark, all 3 points = 2 marks	2
4(a)(ii)	$\text{Na}_2\text{S}_2\text{O}_3$ because reaction stops before all the HCl is used up. OR $\text{Na}_2\text{S}_2\text{O}_3$ because there is still HCl remaining (when the reaction has stopped)	1
4(a)(iii)	the frequency of effective collisions decreases as concentration of HCl(aq) and / or $\text{Na}_2\text{S}_2\text{O}_3$ decreases	1
4(a)(iv)	M1 0.001(00) mol sulfur produced M2 $\text{M1} \times 6.02 \times 10^{23}$ $= 6.02 \times 10^{20}$	2
4(a)(v)	It reacts with water (to produce H_2SO_3)	1

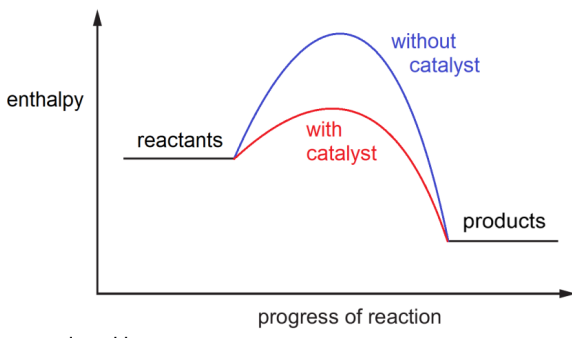
13. M/J 24/P23/Q2(b)(ii)

2(b)(ii)	(increase in temperature / energy from A to C (so)) M1 more collisions with particles that have $E > E_a$ AND increase frequency of effective collisions M2 (initial) rate (of the forward reaction) increases	2
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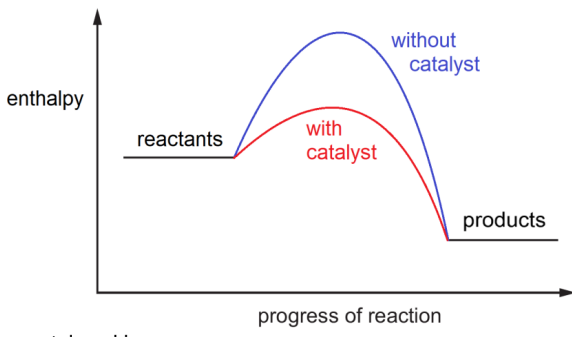
14. MAR 24/P22/Q3(c)(ii)-(iii)

3(c)(ii)	increases rate by providing a different mechanism with a lower activation energy	1
3(c)(iii)	rate is lowered	1
	lower frequency of successful collisions / number of successful collisions per unit time	1

15. O/N 23/P21/Q3(b)

3(b)	 enthalpy reactants without catalyst with catalyst products progress of reaction uncatalysed hump catalysed hump lower than uncatalysed	1
		1

16. O/N 23/P23/Q3(b)

3(b)	 enthalpy reactants without catalyst with catalyst products progress of reaction uncatalysed hump catalysed hump lower than uncatalysed	1
		1

17. M/J 23/P22/Q4(c)(iv)-(v)

4(c)(iv)	M1 a catalyst increases rate of reaction M2 by producing an alternative reaction pathway with lower activation energy	2
4(c)(v)	it is regenerated OR it is reformed	1

18. MAR 23/P22/Q3(d)(i)

3(d)(i)	increases the rate of reaction	1
	by providing an alternative reaction pathway of lower E_a	1

19. M/J 22/P21/Q1(g)

1(g)	M1 (heat / energy released from burning Mg) provides more particles with energy $\geq E_a$ M2 frequency of successful / effective collisions is greater	2
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20. M/J 22/P23/Q3(a)(iv)-(v)

3(a)(iv)	M1 no change / none M2 I_2 is a catalyst AND does not affect position of equilibrium OR I_2 is a catalyst AND reduces time taken to reach (same) equilibrium	2
3(a)(v)	• start at origin and not crossing original until beyond the peak • peak to the right-hand side of original peak AND lower than original • line does not cross the original after the peak reached	2

21. MAR 22/P22/Q2(b)(iii)

2(b)(iii)	two lines shown on diagram, e.g. E_A and $E_{A,cat}$ [1] greater proportion of molecules with $E \geq E_A$ [1] frequency of effective collisions increases [1]	3
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